

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **REST APIs using FastAPI and Flask** for enterprise Java/Spring Boot microservices at HCL, implementing **OOP principles, type hints with Pydantic, and async programming** to handle 500+ concurrent requests; integrated APIs across PostgreSQL, MySQL, and Oracle databases to enable **scalable cloud-native systems** serving banking and ERP clients.
- Built **Python automation pipelines and ETL workflows** using **pandas, NumPy, and Pydantic** for data validation and transformation across enterprise systems; developed **scheduled reporting and data intelligence features** that reduced manual processing time by **60%** and improved data accuracy to **99.8%**.
- Implemented **CI/CD pipelines using Jenkins, GitHub Actions, and ArgoCD** with containerized deployments via **Docker** across AWS (EC2, S3, Lambda, CloudWatch); applied **DevOps practices and cloud-native principles** to achieve **40% faster deployment cycles** and **99.95% application uptime** in production.
- Developed comprehensive **unit and integration tests using PyTest with TDD/BDD practices** and performance benchmarking across REST API endpoints; conducted **code reviews and pair programming sessions** to ensure code quality, reducing production defects by **35%** and improving team knowledge transfer.
- Engineered **security implementations including AuthN/AuthZ using OpenID/OAuth2 standards** and container vulnerability scanning in Docker-based deployments; ensured compliance with **GDPR and regulatory frameworks** across banking production environments with SLA-critical audit trails and change-control management.
- Collaborated across **Agile/Scrum teams with product management, cloud architecture, and data science teams** to deliver iterative outcomes; supported UAT coordination and post-implementation triage for interconnected ERP and business-system integrations, translating technical solutions into measurable business impact.

SKILLS

Backend & API Development – Python, FastAPI, Flask, REST APIs, OOP, Type Hints, Pydantic, Async Programming, Java, Spring Boot.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Containerization, CI/CD Pipelines, Jenkins, GitHub Actions, ArgoCD, Cloud-Native Principles, Linux.

Data Engineering & Analytics – ETL Pipelines, Data Warehousing, pandas, NumPy, scikit-learn, SQL (PostgreSQL, MySQL, Oracle), Data Modelling, Python Data Stack.

Security & Compliance – AuthN/AuthZ, OpenID/OAuth2, Container Vulnerability Scanning, GDPR, Regulatory Compliance, Audit Trails, Change-Control Management.

Testing & Quality Assurance – PyTest, TDD, BDD, Unit Testing, Integration Testing, Performance Testing, Code Reviews, Pair Programming.

Agile & Collaboration – Scrum, Kanban, Jira, Confluence, Cross-functional Teamwork, UAT Support, Stakeholder Communication.

Domain Expertise – Banking & Financial Services, SWIFT Messaging (MT/MX), ISO 20022 Message Flows, SEPA Payments, PSD2, Payments Operations, ERP Integration, Enterprise Business Systems.

EDUCATION AND TRAINING

PROJECTS

AI-Driven Industrial IoT Automation Platform (HCL Technologies)

- Architected and built a **Python-based REST API microservice using FastAPI and Pydantic** to ingest real-time sensor data from 2,000+ connected devices across manufacturing facilities; implemented **async programming patterns and type-safe validation** to process **100K+ events per second** with sub-100ms latency.
- Integrated **AI features for anomaly detection and predictive maintenance automation** using **scikit-learn and pandas** to analyze historical sensor trends and predict equipment failures; deployed models via **AWS Lambda** to trigger automated alerting and downtime reduction of **42%**.
- Containerized the entire platform using **Docker** and orchestrated deployments across **AWS (EC2, S3, CloudWatch)** with **ArgoCD and GitHub Actions CI/CD pipelines**; achieved **rolling blue-green deployments** with zero-downtime releases and **15-minute deployment cycles**.
- Implemented comprehensive **PyTest-based test suites using TDD/BDD practices** covering unit, integration, and performance tests; established **AuthN/AuthZ using OAuth2 standards** and container vulnerability scanning to ensure production-grade security and compliance across multi-tenant industrial customers.
- Collaborated with product management and data science teams to define data schemas, API contracts, and business metrics; delivered iterative releases in **Agile/Scrum sprints** that reduced sensor data processing infrastructure costs by **\$180K annually** through optimized cloud resource utilization.

Real-Time Payment Operations Dashboard & ISO 20022 Integration Platform (HCL Technologies)

- Engineered a **Python FastAPI microservice integrating SWIFT messaging (MT/MX) and ISO 20022 message flow processing** for a Tier-1 banking client; built **REST APIs for payments operations visibility** across SEPA instant payments, cross-border transactions, and transaction reconciliation workflows, handling **50K+ daily payment messages** with SLA compliance of **99.98%**.
- Developed **ETL pipelines using pandas and NumPy** to transform ISO 20022 message structures into operational analytics; created **Power BI dashboards with DAX measures** for real-time payment status tracking, exception reporting, and regulatory audit trail visualization for compliance teams.
- Implemented **Docker containerization and CI/CD automation via Jenkins** to deploy payment-processing services across production environments; applied **GDPR and regulatory compliance frameworks** including encrypted audit logs, role-based access control (OAuth2/OpenID), and change-control workflows for financial-services governance.
- Built **comprehensive PyTest and integration test suites using TDD practices** to validate message parsing, currency conversion, and reconciliation logic; conducted pair programming and code reviews with banking-domain team members, reducing payment-processing defects by **38%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)